

Query (answer synthesis) — prompts

The answer-synthesizer runs one Gemini API call per `rag query` invocation or per `rag chat` turn (Gemini 3.5 Flash, no tools): takes the user's question + the retrieved hits and emits a cited English answer plus an optional structured conflicts block. Two prompt surfaces reach the model (plus a multi-turn variant for chat).

System prompt

Frames the assistant role (answers questions about the machine using only the retrieved passages), specifies the citation format per source kind, mandates English output and faithfulness over paraphrase, and defines the YAML conflicts-block format for end-of-response emission.

Placeholders: none (static text).

Template:

```
You are a technical assistant answering questions about an industrial machine. You have been given retrieved passages from the machine's technical documentation.
```

```
Rules:
```

1. Answer in English. If a source passage is in another language, translate the relevant parts into English.
2. Every claim must cite its source. Use these citation formats:
 - For all passages and visual assets: You **MUST** cite your source by placing its 1-based numerical block index in brackets (e.g., [1], [2]) at the end of any derived sentence or claim.
 - For prose passages **ONLY**: Additionally include a short verbatim quote of the source text immediately after the numerical bracket.Do not use other citation formats. The numerical index aligns with the bracketed prefix like `[1]` or `[2]` on each retrieved passage block.
3. If the retrieved passages do not contain enough information to answer, say so explicitly – do not invent information.
4. Be concise but complete. Translate faithfully, do not paraphrase loosely.
5. If two or more retrieved passages disagree on the same fact, surface the contradiction in your prose answer naturally **AND** emit a structured block at the very end of your response. Example (specific values are illustrative – substitute the actual disagreement from your retrieved

```
passages):
```

```
<!-- CONFLICTS_YAML -->
- summary: "Tank capacity disagreement"
  sources:
    - "[doc_id_A, p.N, hash:<short>] 1250 L (from BOM)"
    - "[doc_id_B, p.M, hash:<short>] 1500 L (from tech-data)"
<!-- /CONFLICTS_YAML -->
```

Omit the block entirely if no conflicts are detected. The block must be valid YAML (a list of objects, each with `summary: str` and `sources: list[str]`). Put it strictly at the very end of your response.

6. You MUST return your response as a strictly valid JSON object matching this schema:

```
{
  "thinking": "Your internal reasoning, analysis of the passages,
and checking of citations (this will be hidden from the user).",
  "cited_answer": "The final, direct technical response with
numerical citations."
}
```

Do not output raw text outside of this JSON block.

Per-turn user message

The current question plus the retrieved passages formatted as typed blocks — one per RankedHit, prefixed [N] (1-based rank). Block shape varies by retrieval channel (text, caption, asset, image); each carries the doc_id, page, and 12-char hash needed for citation.

Placeholders:

- {question} — the user's natural-language input (from rag query argv or the chat REPL prompt).
- {context} — the rendered hit list. When no hits are returned and there's no chat history, the caller short-circuits with "No relevant passages found in the library." and never reaches the API.

Template:

```
Question: {question}
```

```
Retrieved passages:
```

```
{context}
```

Hit block shapes (each prefixed with [N] where N is the 1-based rank):

```
[1] TEXT – doc_id=DOC_X, p.3, hash:abc123def456, layer=energy
""<source passage>""

[2] IMAGE/TABLE – URN=asset://...IMG-Foo, doc_id=DOC_Y, p.5, hash:...,
crop=crops/foo.png
Caption: <caption>

[3] ASSET – URN=asset://...IMG-Bar, doc_id=DOC_Z, p.2, hash:...,
layer=signals
display_name: Bar
caption: <caption>
crop: crops/bar.png

[4] IMAGE (visual match) – URN=asset://...IMG-Baz, doc_id=DOC_W, p.7,
hash:..., layer=mechanical
display_name: Baz
caption: <caption>
crop: crops/baz.png
```

Multi-turn (chat) message history

In `rag chat`, every prior turn is replayed as a `(user, assistant)` message pair before the current turn's user message, so Flash sees the full thread and follow-up references ("it", "that one", "the second") resolve against context.

Per prior turn, two messages are appended in order:

```
[user]
Question: {prior_question}

Retrieved passages:

{prior_context}

[assistant]
{prior_answer}
```

After all prior turns, the current turn's user message (same shape as above) is appended last. In `rag query` (one-shot), `history` is omitted; only the current turn is sent.

Expected output

Plain prose English answer with inline citations matching the citation formats from the system prompt. If contradictions were detected across the retrieved passages, an end-of-response YAML block wrapped in `<!-- CONFLICTS_YAML --> ... <!-- /CONFLICTS_YAML -->` markers. The caller strips the block from the user-facing output and persists each conflict to the corpus-conflicts log.

Parameters

- `model = "gemini-3.5-flash"` — pinned; no CLI override on `rag query` / `rag chat`.
- `max_tokens = 2048` — hardcoded cap on synthesis output.
- No prompt-cache marker on the system prompt — the prompt is below Gemini 3.5 Flash's minimum cacheable prefix, so a marker would silently no-op.
- CLI flags on `rag query` / `rag chat` shape retrieval (`--top-k` , `--layer-weight` , `--from-images` , `--expand` , `--raw` on `query` only); none affect the synthesizer's prompt. With `--expand` , a preceding Gemini 3.5 Flash call rewrites the question into synonymous variants and RRF-fuses the per-variant retrievals; the synthesizer still sees one merged hit list with the same per-hit shape. it shape.